grinding. The setting action starts within five minutes after the addition of water. It becomes hard in less than thirty minutes. This cement is used to lay concrete under static water or running water.

## Rapid Hardening Cement




Rapid Hardening cement attains high hardness in minimum days. The initial setting time and final setting time of this cement is like ordinary cement. Increase in lime content, very fine grinding and burning at high temperature are the reasons for the quick setting. The cost of this cement is more than Portland cement. Construction work is done quickly due to its quick setting and hardening property.

## White Cement




White Cement is a variety of ordinary cement. It is prepared from raw materials which are practically free from oxides of iron, manganese or chromium. It is white in colour. It is used for floor finish, plaster work, ornamental work, etc. It is more costly than ordinary cement.

## Colour Cement






Colour cement is manufactured by mixing 5% to 10% of colouring agents with ordinary cement. Strength of this cement is affected when more than 10% of colouring pigment is added. Green colour is obtained by adding chromium oxide. Blue colour is obtained by adding cobalt. Yellowish brown, red, yellow colours are obtained by adding iron oxide in different proportions. Black colour is obtained by magnesium oxide. This cement is used to make artificial stones, external decoration, coloured cement flooring, etc.

### 4.1.4 Field Tests For Cement

To know the quality of cement, the following tests are conducted in cement.

1. Manufacturing date and Colour test.
2. Physical properties
3. Presence of lumps
4. Block test
5. Glass plate test



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